

3.3.1 Derivatives Worksheets

****Remember that simplifying before attempting to find the derivative is worthwhile****

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions.

a) $y = 4x^3 + e^3$

$$\frac{dy}{dx} = 12x^2 + 0 = 12x^2$$

b) $y = x^{5/3} - x^{2/3}$

$$\frac{dy}{dx} = \frac{5}{3}x^{2/3} - \frac{2}{3}x^{-1/3}$$

c) $A(s) = -\frac{12}{s^5} + \pi^2$

$$A(s) = -12s^{-5} + \pi^2$$

$$\frac{dA}{ds} = 60s^{-6} + 0 = 60s^{-6}$$

d) $g(u) = \frac{u^2 + 4u + 3}{\sqrt{u}}$

$$g(u) = u^{3/2} + 4u^{1/2} + 3u^{-1/2}$$

$$\frac{dg}{du} = \frac{3}{2}u^{1/2} + 2u^{-1/2} - \frac{3}{2}u^{-3/2}$$

e) $w(r) = \sqrt[3]{r}(r-1) + 3e^r$

$$w(r) = r^{4/3} - r^{1/3} + 3e^r$$

$$\frac{dw}{dr} = \frac{4}{3}r^{1/3} - \frac{1}{3}r^{-2/3} + 3e^r$$

f) $h(t) = (t-2)(2t+3)$

$$h(t) = 2t^2 - t - 6$$

$$\frac{dh}{dt} = 4t - 1$$

2. Find the equation of the tangent line to the curve at the given point.

a) $y = x^4 + 2x^2 - x$, $(1, 2)$

We calculate that $y' = 4x^3 + 4x - 1$. We have

Point: $(1, 2)$ & Slope: $m = y'(1) = 7$

Therefore, the equation of the tangent line is $y - 2 = 7(x - 1) \rightarrow y = 7x - 5$.

b) $y = (1 + 2x)^2$, $(1, 9)$

We note that $y = 1 + 4x + 4x^2$. We calculate that $y' = 4 + 8x$. We have

Point: $(1, 9)$ & Slope: $m = y'(1) = 12$

Therefore, the equation of the tangent line is $y - 9 = 12(x - 1) \rightarrow y = 12x - 3$.

c) $y = x + \sqrt{x}$, $(1, 2)$

We calculate that $y' = 1 + \frac{1}{2\sqrt{x}}$. We have

Point: $(1, 2)$ & Slope: $m = y'(1) = \frac{3}{2}$

Therefore, the equation of the tangent line is $y - 2 = \frac{3}{2}(x - 1) \rightarrow y = \frac{3}{2}x + \frac{1}{2}$.

3. Consider the function $y = x^2(1 - 2x) + e$.

a) Determine all value of x at which the function has horizontal tangent lines.

First we note that $y = x^2 - 2x^3$. We calculate that $y' = 2x - 6x^2$.

We want all values of x for which $0 = 2x - 6x^2 \rightarrow 0 = 2x(1 - 3x) \rightarrow x = 0$ or $x = \frac{1}{3}$.

So the function has horizontal tangent lines at x -values of $x = 0$ & $x = \frac{1}{3}$.

b) Determine all value of x at which the function has tangent lines with slope -8 .

We still know that $y' = 2x - 6x^2$. We want all values of x for which

$$-8 = 2x - 6x^2 \rightarrow 0 = -6x^2 + 2x + 8 \rightarrow 0 = -3x^2 + x + 4 \rightarrow 0 = (-3x + 4)(x + 1) \rightarrow x = \frac{4}{3} \text{ or } x = -1.$$

So the function has tangent lines of slope -8 at x -values of $x = -1$ & $x = \frac{4}{3}$.